Machine Learning for Sales Forecasting

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Abstract

Sales forecasting is essential for businesses to optimize inventory and improve business operations. Traditional methods usually aren't robust enough to account for seasonal trends and holiday promotions. This project applies machine learning to predict sales using the Walmart Recruiting dataset, which demonstrates the potential of machine learning for more accurate and robust sales forecasting.

1. Introduction

Accurate sales forecasting is vital for businesses to plan effectively, optimize inventory, and enhance profitability. However, achieving accurate predictions is challenging due to the variety of factors that influence sales patterns, such as holidays, promotions, and regional economic trends. This project tries to address these challenges by utilizing machine learning techniques to improve sales prediction accuracy and reliability.

Traditional sales forecasting methods, such as statistical models and simple trend analysis, often aren't that accurate due to the complexities surrounding sales. These methods struggle to adapt to more complex relationships and variations in the data, especially during critical periods such as holidays or promotional events. Machine learning offers a solution by processing large datasets and uncovering patterns that traditional methods might miss. This project builds on this potential, utilizing the Walmart Recruiting: Store Sales Forecasting Dataset, which provides historical sales data, including information on promotions and holiday markdowns.

1.1. Motivation

Improving sales predictions can make a big difference for businesses. Accurate forecasts help avoid running out of stock, reduce excess inventory, and save money. If businesses can predict sales more effectively, they can make better decisions, satisfy customers, and increase profits. This project aims to show how machine learning can solve this problem and help businesses improve their operations.

1.2. Objectives

The objective of this project is to create and train a model to predict the sales ranges for different businesses in which the model would consider seasonal trends and promotions, leading to a more accurate prediction.

1.3. Current Practices

Currently, businesses most commonly use traditional methods such as linear regression for sales forecasting. Although this method can work for datasets with consistent patterns, it lacks the ability to adapt to other external factors that occur. Things like holidays, the economic state, and promotions aren't considered by a simple linear regression model. Also, when the datasets get too large and complex, it can be extremely difficult for more traditional models to be accurate, leading to missed opportunities for businesses to capitalize on.

1.4. Purpose

If the project is successful and the model can create accurate sales forecasts, then more retail businesses can look at ML to help with inventory management, which can help maximize sales. This will allow businesses to allocate resources more efficiently, plan promotions effectively, and adjust operations to meet customer demand. This improves profitability, strengthens customer loyalty, and reduces operational costs.

2. Approach

In this project, the main objective of creating and training a model that could accurately forecast sales involved several key steps:

```
df_train.describe(include = 'all')
```

	Store	Dept	Date	Weekly_Sales	IsHoliday	Temperature	Fuel_Price	MarkDown1
count	421570.000000	421570.000000	421570	421570.000000	421570	421570.000000	421570.000000	150681.000000
unique	NaN	NaN	143	NaN	2	NaN	NaN	NaN
top	NaN	NaN	2011-12-23	NaN	False	NaN	NaN	NaN
freq	NaN	NaN	3027	NaN	391909	NaN	NaN	NaN
mean	22.200546	44.260317	NaN	15981.258123	NaN	60.090059	3.361027	7246.420196
std	12.785297	30.492054	NaN	22711.183519	NaN	18.447931	0.458515	8291.221345
min	1.000000	1.000000	NaN	-4988.940000	NaN	-2.060000	2.472000	0.270000
25%	11.000000	18.000000	NaN	2079.650000	NaN	46.680000	2.933000	2240.270000
50%	22.000000	37.000000	NaN	7612.030000	NaN	62.090000	3.452000	5347.450000
75%	33.000000	74.000000	NaN	20205.852500	NaN	74.280000	3.738000	9210.900000
max	45.000000	99.000000	NaN	693099.360000	NaN	100.140000	4.468000	88646.760000

Figure 1: Tabular view of training data which shows potential outliers and NaN values (There are more columns in the data frame).

Data Preprocessing:

This dataset is from a previous Kaggle competition from Walmart using their store data. This dataset is extensive with columns such as temperature, fuel price, CPI, unemployment, and store size. I began by just analyzing the dataset and seeing outliers and missing data as shown in Figure 1. Then, I cleaned the dataset to handle missing values and ensure consistency. This step was crucial to prepare the data for effective analysis.

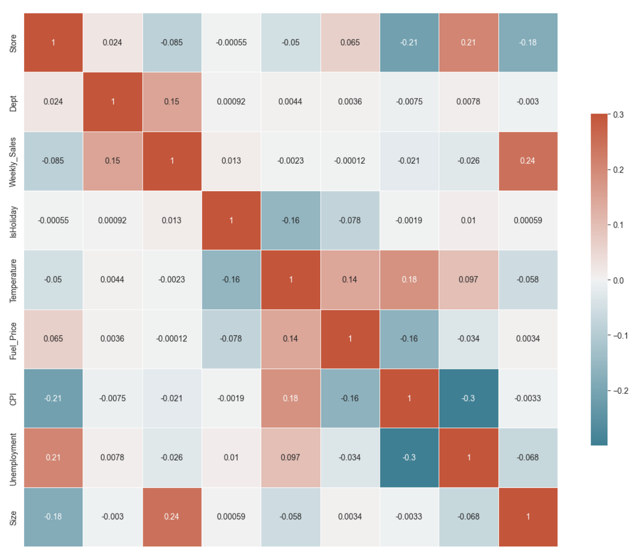

Figure 2: Correlation Matrix to visualize feature correlation for weekly sales.

Feature Engineering:

Feature Engineering is an essential part of this process because it shows which features are the most important and which features have very little impact. As shown in Figure 2, features such as temperature and fuel price can be dropped because their correlation with weekly sales is on the lower side. Additionally, features can be created to capture some more important information. In this case, additional features like the specific week, month, and year as well as holiday type can be added. This holiday-type feature can be further specified on whether there is a holiday and if there is, which one is taking place such as Christmas, Thanksgiving, Labor Day, or the Super Bowl.

$$\text{WMAE} = \frac{1}{\sum w_i} \sum_{i=1}^{n} w_i |y_i - \hat{y_i}|$$

Equation 1: WMAE (Weighted Mean Absolute Error)

Model Selection and Training:

Firstly, the data was split into training and test datasets, so the model's performance was on unseen data. Also, to limit potential overfitting issues, 3-fold cross-validation was done on the model's training process.

Then, the model type had to be selected out of Linear Regression, K-Nearest Neighbors, XGB, and Random Forest. These four options were selected to span a range of complexity and approaches, offering a meaningful basis for comparison:

1. **Linear Regression:** A simple baseline model, providing a reference point for evaluating more complex techniques.
2. **K-Nearest Neighbors:** A memory-based method that can capture local patterns without assuming a specific data distribution.
3. **XGBoost:** A gradient-boosting algorithm known for handling complex relationships and often delivering strong predictive performance.
4. **Random Forest:** A collection of decision trees that balances interpretability, robustness, and accuracy across a variety of problems.

By comparing these models, it can be determined which type of algorithm best fits the data's complexity and ultimately yields the most accurate forecasts.

The evaluation process consisted mainly of the WMAE as shown in Equation 1. The reason behind the WMAE metric is that Walmart outlined the Kaggle Competition as being evaluated by this exact metric.

Evaluation:

A final model was selected from Linear Regression, K-Nearest Neighbors, XGB, and Random Forest.

Hyperparameter optimization was performed based on the model chosen. After that step, the model was trained on the complete dataset and tested with the test dataset.

2.1. Challenges

Initially, I had anticipated that the most difficult parts of this project would come from data processing and model training. Another issue that concerning was the problem of overfitting.

However, the main challenges in this project mainly stemmed from data preprocessing and feature engineering, not so much the model training. The model training was quicker than expected and overfitting was curbed due to the cross-validation process. On the other hand, figuring out how to handle the NaN and missing values was something I had to experiment with. Additionally, figuring out which features to add was also challenging.

3. Experiments

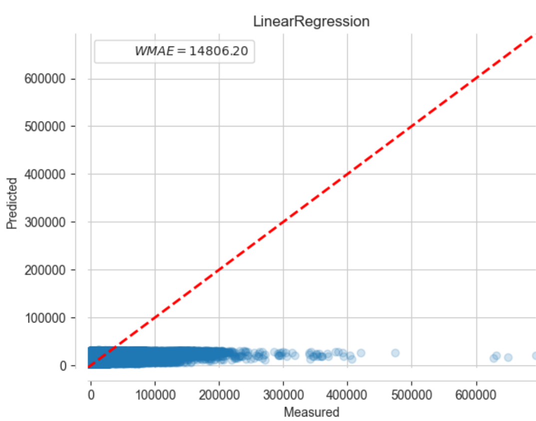

Figure 3: Training evaluation for Linear Regression.

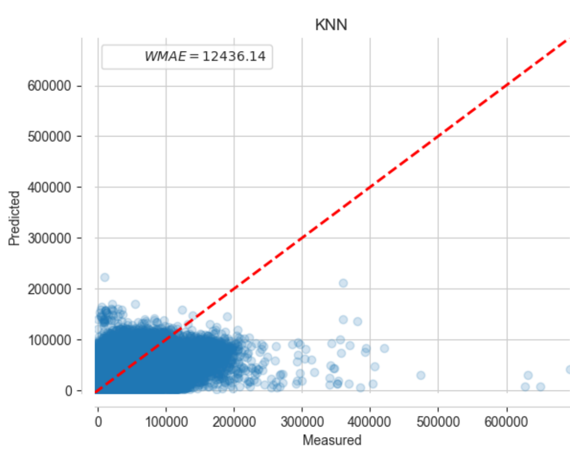

Figure 4: Training evaluation for K-Nearest Neighbors.

Figure 5: Training evaluation for XGB.

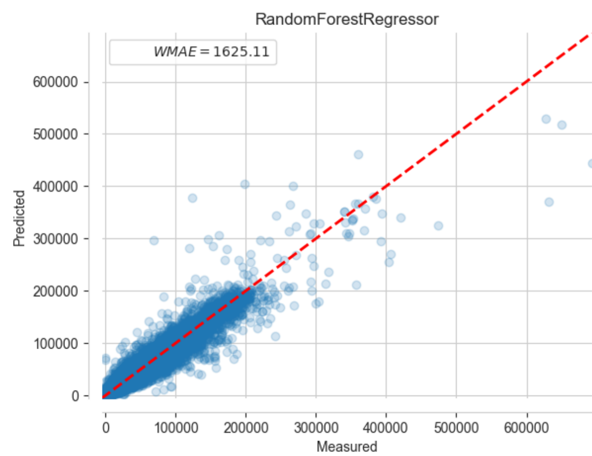

Figure 6: Training evaluation for Random Forest.

The experiment of predicting weekly sales was structured as a supervised regression problem. Each data point consisted of input features and a continuous output target which was the actual sales for that given week. The objective was to learn a mapping from these input features to a numerical sales prediction.

Because the problem is essentially about forecasting a continuous value, models well-suited to regression were chosen, such as Random Forests. These models reflect the structure of the problem by handling a wide array of different features) without requiring extensive manual transformations. They can naturally capture complex interactions between predictors and the target variable, which aligns with the multifaceted nature of sales data influenced by holidays, seasons, and promotions.

3.1 Defining Success

Success was measured using the Weighted Mean Absolute Error (WMAE) metric, which quantifies the average absolute difference between predicted and actual sales values. A lower WMAE indicates higher accuracy.

If a model was obtained that could compete in the 0-2000 WMAE metric range, then it is considered an accurate model with this complex dataset.

3.2 Results

As shown in Figures 3-6, each of the four options: Linear Regression, K-Nearest Neighbors, XGB, and Random Forest are displayed with their training evaluation. This was from Sklearn's framework. This was essential in choosing which model was going to be the final model used in training with the test dataset. Their training metric was the WMAE score explained in Equation 1. As shown in Figure 6, Random Forest had the lowest WMAE score, so it was my choice for the final model. Below in Figure 7-8, are the test performance and feature importance for the Random Forest model.

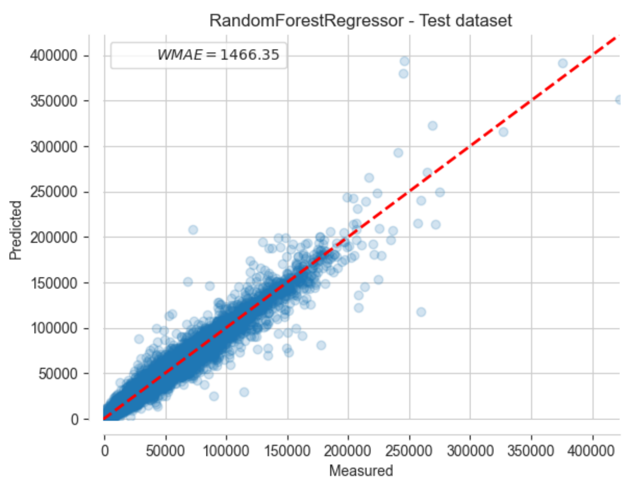

Figure 7: Test evaluation for Random Forest model.

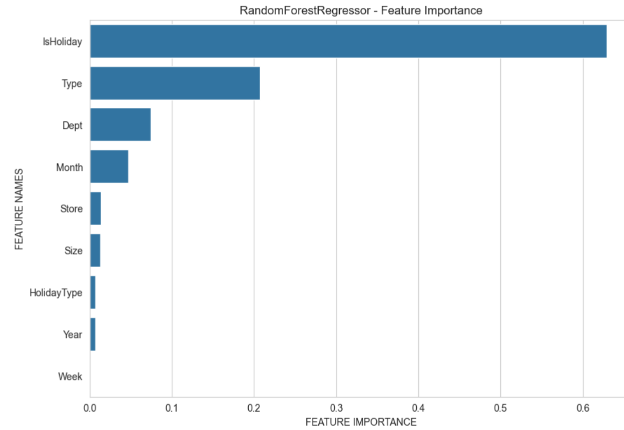

Figure 8: Feature Importance for Random Forest Model.

The project succeeded in developing a predictive model that accurately forecasts weekly sales for Walmart stores. The Random Forest model provided the best performance, because it can capture diverse patterns in the data, leading to improved accuracy. From Figure 8, the holiday feature is a very important feature that is a success because the model takes weeks and weighs them differently depending on if there is a holiday. This robustness allows the model to better adapt to more complex factors such as holidays and seasonal trends.

These findings emphasize the potential of machine learning techniques in improving sales forecasting, offering valuable insights for inventory management and strategic planning.

4. Availability

Code is available on GitHub:
https://github.com/12joshuaslee/MLSalesForecast
Currently, it has an MIT open-source license.

Through this GitHub page, the model is available to be recreated with the same hyperparameters with feature engineering/data preprocessing already completed. The dataset is also on the GitHub, ready to be used.

5. Reproducibility

The Random Forest model can be created similarly but not exactly due to the stochastic nature of the algorithm. It randomly subsamples the data and randomly selects features. However, the graphs and figures should be almost the same. The data is provided through the GitHub link, and I used an 80/20 split for training and test datasets.

6. Conclusion

This project demonstrated that machine learning techniques, especially ensemble models like Random Forest, can improve sales forecasting accuracy over traditional methods. By integrating holiday indicators, temporal features, and careful preprocessing, the chosen model effectively captured complex patterns. As a result, businesses can better anticipate demand, optimize inventory, and make more informed operational decisions. The shared code and datasets facilitate reproducibility and encourage further refinement, offering a strong starting point for organizations seeking to enhance their forecasting strategies.

7. References

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[2] M. Gurnani, Y. Korke, P. Shah, S. Udmale, V. Sambhe and S. Bhirud, "Forecasting of sales by using fusion of machine learning techniques," 2017 International Conference on Data Management, Analytics and Innovation (ICDMAI), Pune, India, 2017, pp. 93-101

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